Three DGP families: five benchmark settings

Write $C = X_2$ for the shared bridge. All root coordinates are mutually independent before selection. Binary roots are uniform on $[-1, 1]$; continuous roots are uniform on $[0, 1]$. $S = 1$ denotes retention for training. Only a joint specification event is removed; individual specification support remains present.

Setting / family	Mechanisms	Training specifications	Held-out test specifications
Weighted Discrete / binary multimodal	$Z_1 = \mathbf{1}\{C \in A,\ \lvert X_1\rvert \le 0.25\}$; $Z_2 = \mathbf{1}\{C \in B,\ \lvert X_3\rvert \le h(C)\}$. $A = [-1, -0.4] \cup [0.4, 1]$, $B = [-0.5, 0.5]$; $h(C) = 0.05$ if $C < 0$, otherwise 0.25.	$(Z_1, Z_2) \in \{(0,0), (1,0), (0,1)\}$; each marginal remains $\{0, 1\}$.	$(Z_1, Z_2) = (1, 1)$ only. Two bridge modes have posterior mass 1/6 (negative) and 5/6 (positive).
Linear / scalar continuous	$Z_1 = 0.8X_1 + 0.6C$; $Z_2 = 0.8X_3 + 0.6C$.	Each marginal support: $[0, 1.4]$; joint support $\mathcal{S}_Z \setminus \mathcal{R}$.	$I_1 = [0.3135, 0.3165]$; $I_2 = [1.0835, 1.0865]$.
Cubic / scalar continuous	$Z_1 = 0.8X_1 + 0.6C$; $Z_2 = 0.8X_3 + 0.6C^3$.	Each marginal support: $[0, 1.4]$; joint support $\mathcal{S}_Z \setminus \mathcal{R}$.	$I_1 = [0.3135, 0.3165]$; $I_2 = [0.862803125, 0.865803125]$.
Smoothstep / scalar continuous	$Z_1 = 0.8X_1 + 0.6C$; $Z_2 = 0.8X_3 + 0.6(3C^2 - 2C^3)$.	Each marginal support: $[0, 1.4]$; joint support $\mathcal{S}_Z \setminus \mathcal{R}$.	$I_1 = [0.3135, 0.3165]$; $I_2 = [1.07601875, 1.07901875]$.
2D Bridge / vector continuous	$C = (u, v)$; $Z_1 = 0.7u + 0.3v + (0.8 + 0.15uv)X_1$; $Z_2 = 1 - 0.3u - 0.7v +$ $[0.8 + 0.15(1-u)(1-v)]X_3$.	Each marginal support: $[0, 1.95]$; joint support $\mathcal{S}_Z \setminus \mathcal{R}$.	$I_1 = I_2 = [0.3485, 0.3515]$.

Table 1: Exact implemented mechanisms and split definitions. For continuous settings, $\mathcal{R} = I_1 \times I_2$ is the excluded/test rectangle and $\mathcal{S}_Z$ is the mechanism-induced joint support. Training is not uniform over the displayed marginal ranges or their Cartesian product. All rectangle side lengths are 0.003; there is no additional exclusion buffer.

Sample counts. Each seed draws a 200,000-root candidate pool, then selects 20,000 retained training rows and 2,000 retained validation rows. There are three seeds. Binary testing uses 2,000 conditional reference samples at $(1, 1)$. Continuous testing uses 64 held-out target pairs, with 32 reference draws per target (2,048 outputs). Continuous targets are drawn from the population conditional on the held-out rectangle; they are not assumed uniform in that rectangle.

Bridge compatibility and selection audit

For a fixed target z, define $F_i(z_i) = \{c : \exists x_{\text{private}} \text{ in root support with } f_i(x_{\text{private}}, c) = z_i\}$. Thus compatibility means that a private root *can* satisfy the specification; it is not sufficient for an arbitrary generated private root to satisfy it.

Setting	Bridge satisfying Z_1 alone	Bridge satisfying Z_2 alone	Bridge satisfying both	Center measure reduction
Weighted Discrete	$[-1, -0.4] \cup [0.4, 1]$	$[-0.5, 0.5]$	$[-0.5, -0.4] \cup [0.4, 0.5]$	Lengths $1.2, 1.0 \to 0.2$; ratio 0.20.
Linear	$[0, 0.525]$	$[0.475, 1]$	$[0.475, 0.525]$	$0.525, 0.525 \to 0.05$; ratio 0.09524.
Cubic	$[0, 0.525]$	$[0.475, 1]$	$[0.475, 0.525]$	Same center lengths.
Smoothstep	$[0, 0.525]$	$[0.475, 1]$	$[0.475, 0.525]$	Same center lengths.
2D Bridge	$\{(u, v) \in [0, 1]^2 : \ 0.7u + 0.3v \leq 0.35\}$	$\{(u, v) \in [0, 1]^2 : \ 0.3u + 0.7v \geq 0.65\}$	Intersection of the two polygons. Coordinate projections: $u \in [0, 0.125]$, $v \in [0.875, 1]$; **not their Cartesian product**.	Areas $2/7, 2/7 \to 1/112$; ratio $1/32 = 0.03125$.

Table 2: Exact compatibility sets at the test-rectangle center (binary target $(1, 1)$). The scalar continuous centers were chosen to share the same bridge overlap, isolating changes in mechanism. The ratio is joint measure divided by the smaller individual measure.

Away from the center. For the three scalar continuous mechanisms, let $g(c) = c$, c^3, or $3c^2 - 2c^3$. Throughout the specified test rectangle,

$$F_1(z_1) = [0, z_1/0.6], \qquad F_2(z_2) = [g^{-1}((z_2 - 0.8)/0.6), 1], \qquad F_1 \cap F_2 = [g^{-1}((z_2 - 0.8)/0.6), z_1/0.6].$$

The joint width ranges are $[0.045, 0.055]$ (Linear), approximately $[0.043835, 0.056223]$ (Cubic), and $[0.045829, 0.054171]$ (Smoothstep). For 2D, replace 0.35 by z_1 and 0.65 by $1 - z_2$ in the inequalities; the audited joint area varies approximately from 0.00840 to 0.00947.

Setting	$P(S = 0 \mid E_1)$	$P(S = 0 \mid E_2)$	$P(S = 0)$	Threshold	Assessment
Weighted Discrete	2.5%	5.0%	0.375%	1%	Relaxed reference; does not pass
Linear	0.0357143%	0.0357143%	0.0000703124%	1%	Passes band-conditional audit
Cubic	0.0357232%	0.0357224%	0.0000703218%	1%	Passes band-conditional audit
Smoothstep	0.0357152%	0.0357151%	0.0000703089%	1%	Passes band-conditional audit
2D Bridge	0.0116082%	0.0116082%	0.0000123245%	1%	Passes band-conditional audit

Table 3: Operational Assumption-3 exclusion audit, not a binary assertion that the theoretical assumption holds exactly. $E_i = \{Z_i = 1\}$ for binary data and $E_i = \{Z_i \in I_i\}$ for continuous data. Conditional continuous rates use the saved refined quadrature; overall rates use the original saved quadrature.

These continuous conditional rates are **band averages**, not verified bounds on $\sup_{z_i \in I_i} P(S = 0 \mid Z_i = z_i)$. Root independence holds before selection; selection can distort the observed factors. Small exclusion and retained marginal support do not prove an undistorted likelihood or accurate learned models. Geometric compatibility above uses exact target equations; the result evaluator separately allows $|f_i(X) - z_i| \leq 0.005$ on each continuous specification and also requires root support.